

The Brain-Pong Starter Protocol

An 8-Week, Evidence-Based Introduction to Table Tennis for Aging Brains and Bodies

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What This Is

This protocol turns peer-reviewed clinical research into a practical 8-week plan for adults 55 and older who want to use table tennis to protect balance, cognition, and cardiovascular health. Every claim in these pages traces to a published study, cited at the end.

It is written for two audiences at once:

- **Individuals** using it on their own or with a partner at a club, community center, or home table.
- **Activity and rehab professionals** running group sessions in senior residences, day programs, and clinics. Facility notes appear in boxed sections like this one.

Facility note: A licensed facility edition with printable session cards, staff scripts, and progress tracking sheets is available. See the last page.

What the Research Shows (In Brief)

Four findings anchor this protocol:

1. **Cognition improves dramatically.** A 2026 meta-analysis of 14 randomized controlled trials and 1,565 participants found table tennis improved cognitive function with a standardized mean difference of 2.05. For context, 0.8 is considered a large effect in behavioral science. Balance improved at 0.78, a medium-to-large effect. Longer programs produced larger cognitive gains. (Li, Ahn & Shin, Healthcare, 2026)
2. **Aging bodies respond within 12 weeks.** A randomized trial in adults aged 55 to 65 found 12 weeks of table tennis, three sessions per week, improved balance, reaction time, grip

strength, body composition, and antioxidant defenses. Participants were beginners. (Scientific Reports, 2025)

3. **Parkinson's patients improved on every measured outcome.** A 12-week structured program called NeuroPong, twice weekly for 2 hours, improved motor scores by roughly 18%, plus balance, walking endurance, and depression, with 90% attendance and no adverse events. (Barbera et al., NeuroRehabilitation, 2026)
4. **Open-skill exercise protects the aging brain better than repetitive exercise.** Reviews in Brain Sciences conclude that activities requiring constant reaction to an unpredictable opponent, like table tennis, outperform closed-skill activities such as treadmill walking for preventing cognitive decline. (Kumagai et al., 2025)

Who This Protocol Suits

- Healthy adults 55+ wanting a scientifically supported way to train balance and cognition
- Adults with mild cognitive impairment or early Parkinson's, with physician clearance
- Cardiac rehab graduates seeking an engaging step-down activity (see week 5 onward and the safety section)

Who should get medical clearance first: anyone with uncontrolled hypertension, recent cardiac events, significant osteoporosis, untreated cataracts or severe vision loss, or a history of falls in the last 6 months. Bring this document to your appointment. The intensity is low to moderate, but clearance is cheap and falls are not.

What You Need

- Access to any table tennis table (club, community center, residence common room)
- A paddle and at least one box of 40+ mm training balls
- Comfortable athletic shoes with grip (not running shoes with worn soles)
- A partner, or a group of 3 to 6 rotating players, or a coach for multiball drills
- A chair within reach of the table at all times

Optional but valuable: a ball catcher net that mounts on the table edge, reducing pickup bending, and a robot feeder if practicing alone (robots cannot replace the cognitive challenge of a live opponent, which is where most of the brain benefit comes from).

The 8-Week Protocol

Built from the session structures used in the trials above: the 55 to 65 trial used 3 sessions per week, the Parkinson's program used 2, the cardiac program used 3 with multiball intervals. This protocol defaults to **2 sessions per week, 60 minutes each**, the gentlest evidence-based dose, with an optional third session marked where the research supports it.

The structure of every session, mirroring the trials:

1. **Warm-up, 10 minutes:** seated marching, shoulder circles, side steps along the table length, gentle torso rotations
2. **Skill block, 25 minutes:** the week's drills, at the intensity described
3. **Game block, 20 minutes:** structured rally play, scoring optional
4. **Cool-down, 5 minutes:** walking, stretching calves, shoulders, wrists

Weeks 1 and 2: Control and Contact

Goal: ball familiarity, safe stance, controlled rally of 5+ shots.

Drills: - **Ball familiarization (10 min):** bounce the ball on the paddle while standing, then walking. Then alternate forehand and backhand bounces. This trains the wrist and eye-tracking gently. - **Shadow strokes (5 min):** practice forehand and backhand strokes without a ball, focusing on weight shift from back foot to front foot. Cue: "hand moves because feet move." - **Blocked forehand (10 min):** partner or coach feeds by hand, you return forehand only, no scoring. Aim for consistency, not speed. - **Rally target game (10 min):** count consecutive shots with your partner. Try to beat your own record from the previous session.

Intensity: low. You should be able to speak in full sentences throughout.

Facility note: With 4 or more participants, run blocked feeding in pairs while others do ball familiarization seated, then rotate every 10 minutes. One staff member can run a group of 8.

Weeks 3 and 4: Footwork and Prediction

Goal: side-stepping to the ball, 10+ shot rallies, first multiball exposure.

Drills: - **Side-step touch (5 min)**: side-step the width of the table, touching the far corner, return. Repeat at comfortable pace. - **Two-point forehand (10 min)**: partner feeds alternately to your forehand corner and middle; you side-step between shots. - **Multiball introduction (10 min)**: a feeder drops balls in a steady rhythm to your forehand. Return each without rushing. The cardiac rehab trial used multiball in 5-minute blocks, one group every 5 minutes. Start with two blocks. - **Point play (15 min)**: play games to 5 points, serving underhand from an open palm.

Intensity: low to moderate. By the end of week 4, elevated breathing during multiball is expected and appropriate.

Weeks 5 and 6: Reaction and Decision

Goal: unpredictable placement, which is where the cognitive load lives.

Drills: - **Random placement (10 min)**: partner feeds to forehand or backhand without a pattern. You decide and move. This unpredictability is the "open-skill" element the brain research highlights. - **Multiball, mixed (10 min)**: feeder varies depth and side in a light rhythm. - **Constraint games (15 min)**: play points where one player may only hit to the other's backhand, or must announce "now" before switching sides mid-rally. Constraints force planning and inhibition, the executive functions the trials measured. - **Optional third session**: 30 minutes of rally play only.

Intensity: moderate. Brief breathlessness after long rallies is fine. Chest pain, dizziness, or joint pain means stop and note it for your physician.

Weeks 7 and 8: Integration

Goal: full games, sustained rallies of 15+ shots, measurable progress.

Drills: - **Serve and third ball (10 min)**: serve, partner returns, you attack the return once, then rally continues. Trains sequencing. - **Match play (25 min)**: games to 11, rotating opponents if in a group. - **Benchmark re-test (final session)**: (a) consecutive rally count vs. your week 1 count, (b) side-step touch: laps of the table width in 30 seconds vs. week 1, (c) single-leg stance time on each leg vs. week 1. Write them down. Improvement on all three is typical; the 55 to 65 trial saw reaction-time and grip improvements in 12 weeks at this dose.

Where to go after week 8: the meta-analysis found cognitive benefits grow with program length, and the cardiac trial ran 4 months. The natural progression is to keep 2 sessions per week indefinitely, join a club league if competitive play appeals, and re-run the benchmarks every 8 weeks.

Safety Notes

- Keep a chair at courtside. Sit when tired. The trials allowed rest freely and still saw gains.
- Hydrate before and after each session.
- Stop immediately for chest pain, pressure, dizziness, or unusual breathlessness at rest, and seek medical care.
- Ankle and knee injuries are the main risks in racket sports; table tennis has the lowest injury rate among them (a 2024 systematic review of 873 professional players found a 3.6% injury rate, most causing no lost time). Correct footwear and the side-step technique in week 3 are your protection.
- For Parkinson's participants: freezing of gait can occur during quick direction changes. The NeuroPong program used large visual targets and allowed extra time between drills. Assign a standing partner during footwork blocks.
- For cardiac rehab graduates: the evidence comes from a trial where table tennis was added to standard resistance rehab under supervision. Treat weeks 1 to 4 as your ramp and clear the protocol with your cardiologist first.

Progress Tracking Sheet

Photocopy or reprint one per 8-week cycle.

Week	Sessions done	Longest rally	Notes (energy, aches, mood)
1	/2		
2	/2		
3	/2		
4	/2		
5	/2		
6	/2		
7	/2		
8	/2		

Benchmarks: single-leg stance (L/R): week 1 **_ s / s**, week 8 **_ s / s**. Side-step laps in 30 s: week 1 **_**, **week 8 _**.

References

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2. Scientific Reports (Nature). 12-week randomized controlled trial in adults 55-65. 2025. DOI 10.1038/s41598-025-31098-8. <https://www.nature.com/articles/s41598-025-31098-8>
3. Barbera A, Woodward M, Buard I, Sillau S, Montes RO, Feuerstein J. A pilot table tennis-based neurorehabilitation program to improve Parkinson's disease symptoms. *NeuroRehabilitation*. 2026. DOI 10.1177/10538135261468787. PMID 42454977.
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7. Wei JN, Zhang MK, Wang Z, Liu Y, Zhang J. Table tennis experience enhances motor control in older adults. *International Journal of Clinical and Health Psychology*. 2024;24(2):100464. DOI 10.1016/j.ijchp.2024.100464.
8. Sherrington C, et al. Exercise for preventing falls in older people living in the community: an umbrella review. *International Journal of Behavioral Nutrition and Physical Activity*. 2020. DOI 10.1186/s12966-020-01041-3.

Disclaimer: This document summarizes published research and provides general exercise guidance. It is not medical advice. Consult a physician before beginning this or any exercise program, particularly with existing cardiac, neurological, or orthopedic conditions.

Facility Edition: residences, day programs, and clinics can license the full pack (session cards, staff scripts, assessment forms, progress sheets for unlimited participants, and a staff training overview). Contact licensing@pongnews.com.